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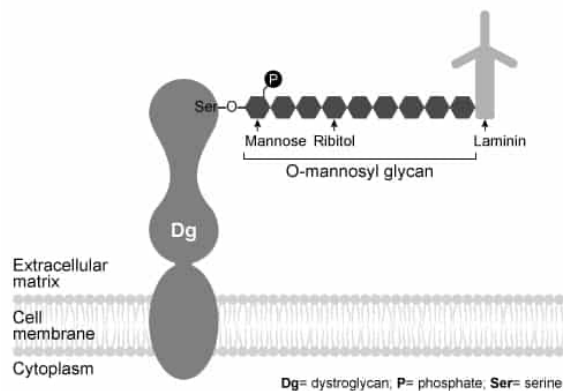


Figure 1 Schematic of interaction between Dg and laminin, mediated by the Dg-linked O-mannosyl glycan

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Researchers investigating dystroglycanopathies recently discovered the functions of two enzymes in the pathway: protein O-mannose kinase (POMK) and fukutin. POMK phosphorylates Dg-linked mannose at carbon 6, and fukutin adds the sugar ribitol 5-phosphate to the glycan chain. To assess the roles of POMK and fukutin relative to each other, scientists conducted experiments in which lysates from muscle cells deficient in either POMK or fukutin were incubated with ^{32}P -labeled ATP or ^3H -labeled activated ribitol 5-phosphate. Combined lysates containing both fukutin and POMK were also tested. Dg was then purified and detected by autoradiography and western blot using an antibody against Dg, as shown in Figure 2.

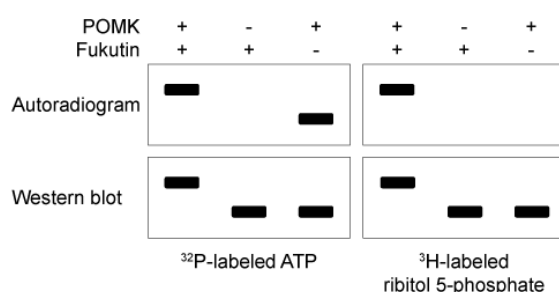
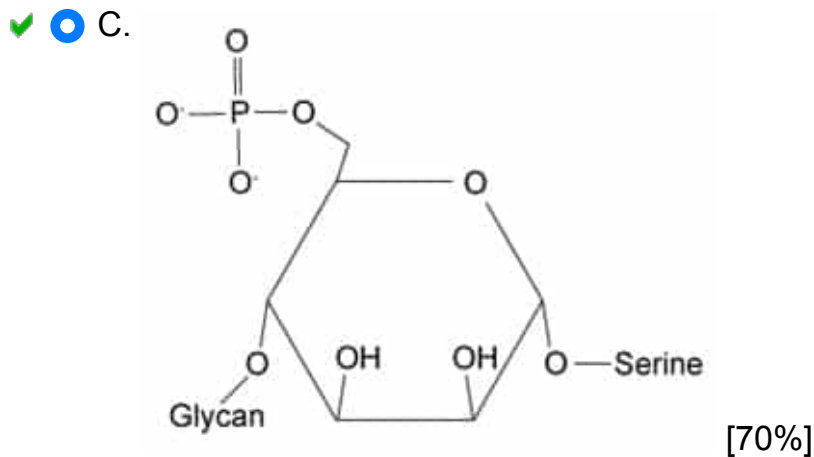
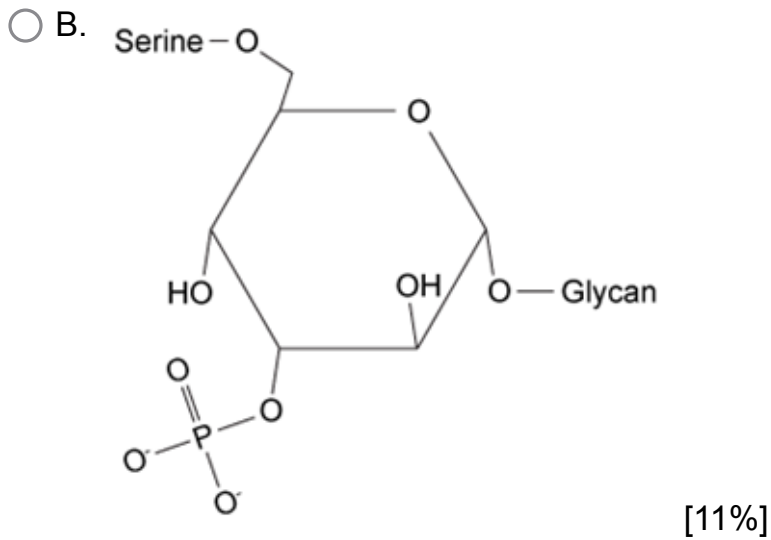
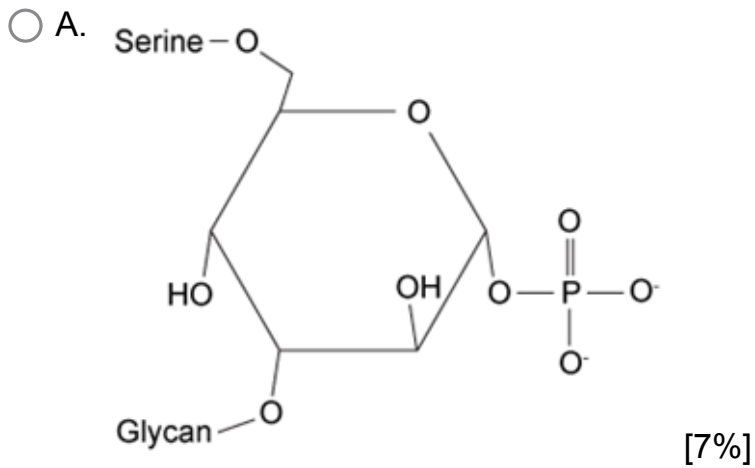
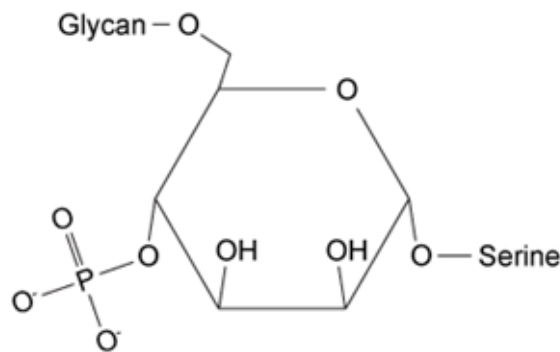


Figure 2 Autoradiograms and western blots of Dg purified from lysates incubated with ^{32}P -labeled ATP or ^3H -labeled ribitol 5-phosphate

Which of the following structures could represent the completed O-mannosyl glycan?



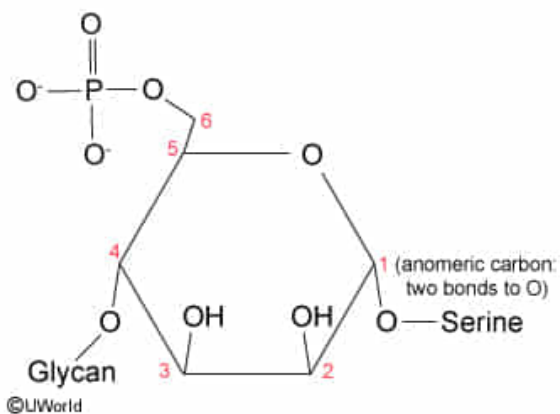


[9%]

Correct

70% answered correctly

Explanation:



Carbohydrate **carbons are numbered** starting at the end that is closest to the **anomeric carbon**, which is the one with two bonds to oxygen. In the case of **aldoses** such as **D-mannose**, the anomeric carbon is carbon 1 whereas in **ketoses** such as **D-fructose**, the anomeric carbon is typically carbon 2. Most carbohydrates can exist in **both linear and cyclical forms**, and in cyclical form the anomeric carbon can adopt one of two **conformations**, known as the **α - and β -conformations**.

The passage states that D-mannose connects to serine through an **α -linkage** indicating that the linkage is through the anomeric carbon (carbon 1) in the α -conformation. D-mannose connects to the next sugar in the chain in a **β -1,4 linkage**, which indicates a bond between the anomeric carbon of a sugar in the β -conformation and carbon 4 of another sugar. Because the anomeric carbon of D-mannose is already linked to serine, carbon 4 of D-mannose must be linked to the anomeric carbon of the next sugar in the chain. Finally, POMK attaches a phosphate group to D-mannose at carbon 6.

(Choices A and B) These choices show serine linked to carbon 6, which is not the anomeric carbon. Furthermore, the oxygen on carbon 3 is shown below the plane of the ring, making the sugar D-altrose instead of D-mannose.

(Choice D) This option switches the positions of the phosphate group and the glycan chain.

Educational objective:

Carbohydrate carbons are numbered starting at the end of the linear chain closest to the anomeric carbon. Most sugars can exist as linear or cyclical molecules and upon cyclization can adopt either an α - or a β -conformation at the anomeric carbon.

Time spent:0 sec**Subject:**

Biochemistry

QID:401004**System:**

Carbohydrates, Nucleotides, and Lipids

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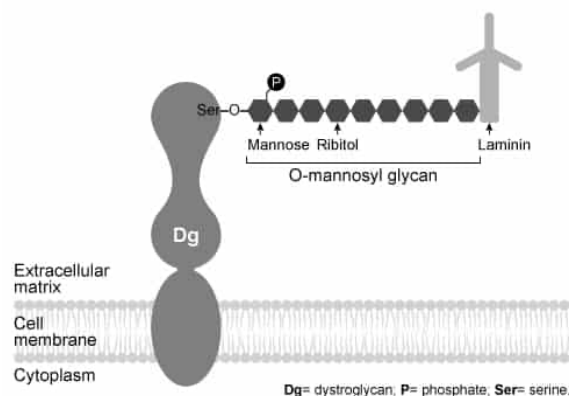


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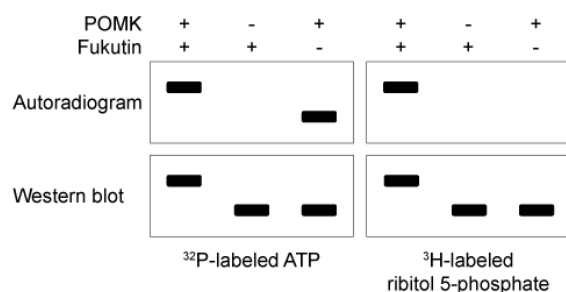


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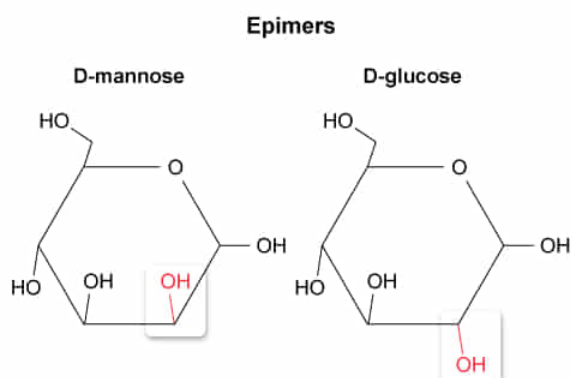
POMT can use activated D-mannose, but not D-glucose, as a substrate. D-glucose and D-mannose are:

- ☐ A. constitutional isomers. [17%]
- ☐ B. enantiomers. [6%]
- ✓ ☒ C. epimers. [68%]
- ☐ D. anomers. [7%]

Correct

67% answered correctly

Explanation:



Epimers have multiple stereocenters but differ only at one.

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Molecules with the same molecular formula but different arrangements of the atoms are called isomers. **Stereoisomers** are molecules with the same kinds of bonds at any given position but with the bonds **oriented differently in space**. Stereoisomers have one or more **stereocenters**, also called chiral centers, which are atoms bonded to four different chemical groups. Molecules that have **multiple stereocenters** but **differ at only one of them** are called **epimers**.

D-mannose and D-glucose each have several stereocenters but differ from each other only at the carbon 2 center; therefore, they are called C-2 epimers. Because POMT can use activated D-mannose but not D-glucose as a substrate, it must be able to distinguish between these epimers.

(Choice A) Unlike stereoisomers, **constitutional isomers** have the same molecular formula but different bond types in at least one position (ie, different connectivity between atoms). For example, **D-glucose and D-fructose** are constitutional isomers, having a carbon with two bonds to oxygen at carbon 1 and carbon 2, respectively.

(Choice B) **Enantiomers** are stereoisomers that differ at every stereocenter, not just

one. The **enantiomer** of D-mannose is L-mannose.

(Choice D) Anomers are a special kind of epimer that differ at the anomeric carbon and are designated as the **α - and β -forms** of the same sugar. This difference can only occur in the cyclical form of the carbohydrate because the anomeric carbon is not a stereocenter in the linear form.

Educational objective:

Stereoisomers are molecules that differ from each other in the spatial orientation of their bonds at one or more stereocenters (atoms that are bonded to four different chemical groups). Stereoisomers that differ at only one of several centers are epimers whereas those that differ at all centers are enantiomers. Constitutional isomers have the same molecular formula but distinct bond arrangements.

Time spent:0 sec

Subject:
Biochemistry

QID:401005

System:
Carbohydrates, Nucleotides, and Lipids

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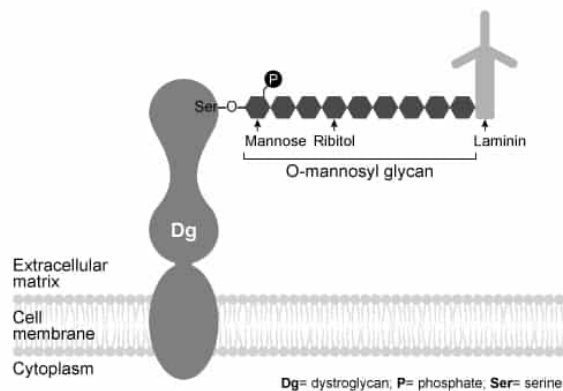


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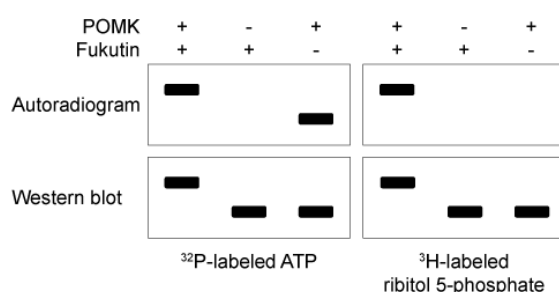


Figure 2 Autoradiograms and western blots of Dg purified from lysates incubated with ^{32}P -labeled ATP or ^3H -labeled ribitol 5-phosphate

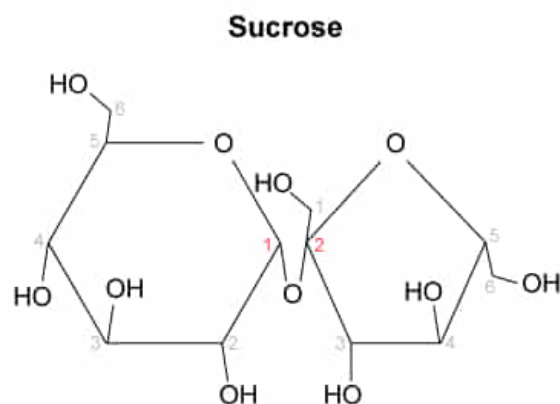
O-linked D-mannose can form a glycosidic bond with any sugar that has a free anomeric carbon. Therefore it could theoretically form a glycosidic bond with all of the following sugars EXCEPT:

- ☐ A. galactose. [6%]
- ☐ B. ribose. [28%]
- ☐ C. lactose. [14%]
- ✓ ☒ D. sucrose. [49%]

Correct

49% answered correctly

Explanation:



Linked anomeric carbons (reducing ends) can't participate in additional glycosidic bonds.

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A glycosidic bond is a bond between the anomeric carbon of a carbohydrate and any other biological molecule. The **formation of glycosidic bonds** requires at least one sugar with a **free anomeric carbon**, also known as a **reducing end**. A carbohydrate with a free reducing end is called a **reducing sugar**.

Because O-linked D-mannose is connected to Dg through its anomeric carbon, it does not have a free reducing end and will only be able to form a glycosidic bond with a sugar that does have an available reducing end. Sucrose is a disaccharide in which the reducing end of D-glucose is linked to the reducing end of D-fructose. Because both reducing ends are already involved in a glycosidic bond, sucrose is not a reducing sugar. Therefore, O-linked mannose will not be able to form a glycosidic bond with sucrose.

(Choices A and B) Galactose and ribose are monosaccharides (single carbohydrate units). All free monosaccharides are reducing sugars and can form glycosidic bonds.

(Choice C) Lactose is a disaccharide formed by a β -1,4 linkage between galactose and glucose. The glucose unit has a free anomeric carbon, and therefore is a reducing sugar.

Educational objective:

Glycosidic bond formation requires a sugar with a free anomeric carbon, called a reducing sugar. All free monosaccharides are reducing sugars whereas disaccharides are reducing only if one anomeric carbon does not participate in a glycosidic bond. Sucrose is the most common nonreducing sugar.

Time spent:0 sec

Subject:

Biochemistry

QID:401006

System:

Carbohydrates, Nucleotides, and Lipids

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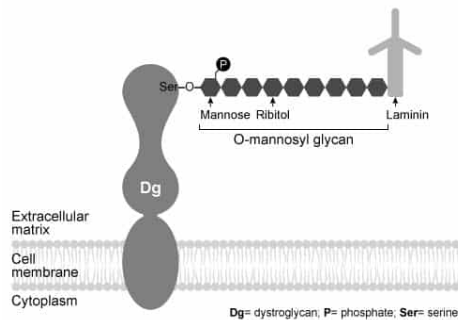


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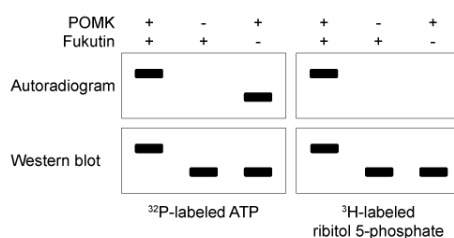


Figure 2 Autoradiograms and western blots of Dg purified from lysates incubated with ^{32}P -labeled ATP or ^3H -labeled ribitol 5-phosphate

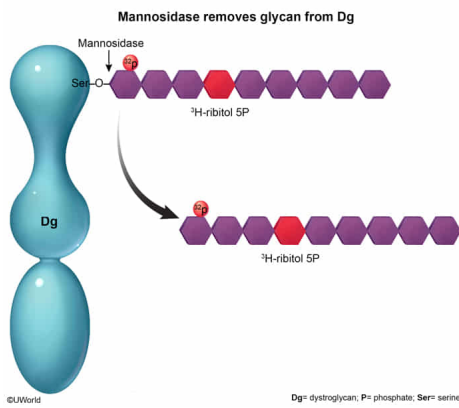
Alpha mannosidase cleaves the link between D-mannose and the hydroxyl group of serine or threonine. If researchers treated the radiolabeled Dg from each autoradiogram with alpha mannosidase, where would the radiolabels be found?

- ☐ A. Both ^{32}P and ^3H -ribitol would be attached to the Dg protein. [7%]
- ☐ B. The ^{32}P would be attached to Dg and the ^3H -ribitol to the cleaved polysaccharide. [28%]
- ☐ C. The ^3H -ribitol would be attached to Dg and the ^{32}P label to the cleaved polysaccharide. [25%]
- ☒ D. Both ^{32}P and ^3H -ribitol would be found in the cleaved polysaccharide. [38%]

Correct

37% answered correctly

Explanation:



According to the passage, D-mannose is the first sugar in the glycan chain and links to Dg via a glycosidic bond through the anomeric carbon of D-mannose. Mannosidase is in a class of enzymes called **glycosidases** that **break glycosidic bonds**. In this case, the broken bond is the bond that links the glycan chain to Dg through D-mannose, so the entire glycan chain will be removed from Dg as a single polysaccharide. The ^{32}P is attached to carbon 6 of mannose, and the ^3H ribitol 5-phosphate is within the cleaved polysaccharide. Therefore, **both radiolabels** will be removed from the Dg protein and **remain** on the **cleaved polysaccharide**.

(Choice A) The polysaccharide would have to be cleaved between the ribitol 5-phosphate added by fukutin and the end of the glycan chain for both radiolabels to remain associated with Dg. Instead, mannosidase cleaves the bond that links D-mannose to Dg and removes all radiolables from Dg.

(Choice B) If the polysaccharide were cleaved between mannose and ribitol 5-phosphate, this would be the correct answer. Because the bond linking the polysaccharide to Dg is cleaved, both radiolabels are removed from Dg.

(Choice C) For this option to be correct, the polysaccharide would have to be cleaved in such a way as to remove mannose from Dg without removing ribitol 5-phosphate. Because mannose is the only sugar directly linked to Dg, this is not possible.

Educational objective:

Glycosidases break glycosidic bonds to produce monosaccharides, disaccharides, and polysaccharides. The length and composition of the newly formed saccharides depends on which glycosidic bond the glycosidase breaks.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401007

System:
Carbohydrates, Nucleotides, and Lipids

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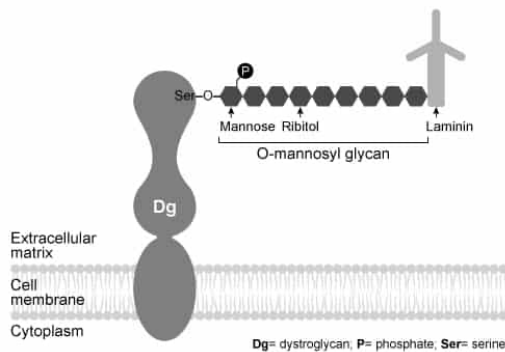


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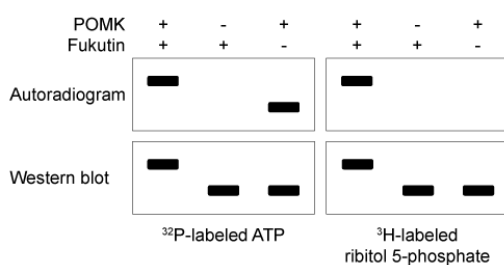


Figure 2 Autoradiograms and western blots of Dg purified from lysates incubated with ^{32}P -labeled ATP or ^3H -labeled ribitol 5-phosphate

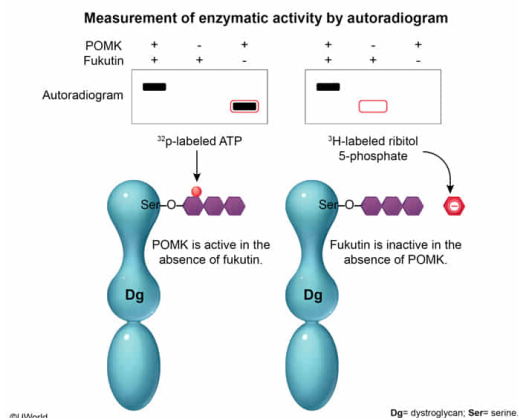
Which conclusion about the functions of POMK and fukutin is supported by the autoradiogram in Figure 2?

- ☐ A. Fukutin activity must occur before POMK activity can occur. [6%]
- ✓ ☒ B. POMK activity must occur before fukutin activity can occur. [65%]
- ☐ C. Fukutin and POMK activities can occur in any order. [12%]
- ☐ D. POMK and fukutin activities must occur simultaneously. [16%]

Correct

65% answered correctly

Explanation:



In **autoradiography**, potentially radiolabeled biomolecules, including proteins, nucleic acids, and carbohydrates, are separated in gels. **Radioactivity is then detected** by exposure to radiation-sensitive film, and molecules that contain radioactive atoms such as ^{32}P or ^3H show up as bands. Molecules without radioactive atoms do not produce visible bands.

Figure 2 shows that ^{32}P from labeled ATP is incorporated into Dg in all cases tested except when POMK is absent. POMK is a kinase, meaning that it transfers a phosphate from ATP to a target molecule, so the lack of ^{32}P -incorporation in the absence of POMK is expected. Similarly, ^3H from ribitol 5-phosphate is not incorporated when fukutin is absent, which is expected because fukutin attaches ribitol 5-phosphate to the growing glycan chain. However, the figure also shows that in the presence of fukutin, ribitol 5-phosphate is incorporated into Dg when POMK is present but not when POMK is absent. Therefore, **POMK activity must occur before fukutin** can catalyze the addition of ribitol 5-phosphate to the glycan, and must occur first.

(Choice A) If fukutin activity needed to occur before POMK activity could, no ^{32}P -labeled band would appear in the lane where fukutin is absent, and a ^3H -labeled band would appear in the lane where POMK is absent but fukutin is present.

(Choice C) If fukutin and POMK activities could occur in any order, fukutin would be active in the second lane of the ^3H autoradiogram, and a band would be visible.

(Choice D) If POMK and fukutin activities needed to occur simultaneously, neither would show activity in the absence of the other and only the lanes with both enzymes would have visible bands in the autoradiogram.

Educational objective:

Autoradiography can detect the incorporation of radioactive atoms into biomolecules. It can help determine the conditions under which an enzyme can act on its substrate.

Time spent:0 sec

Subject:

Biochemistry

QID:401008

System:

Carbohydrates, Nucleotides, and Lipids

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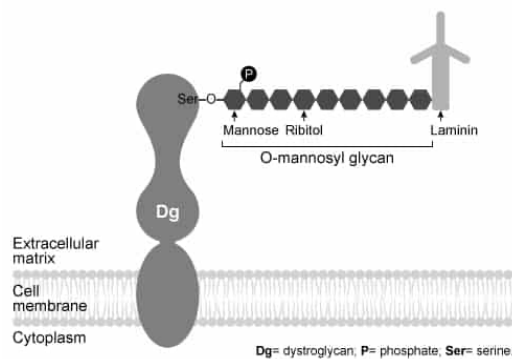


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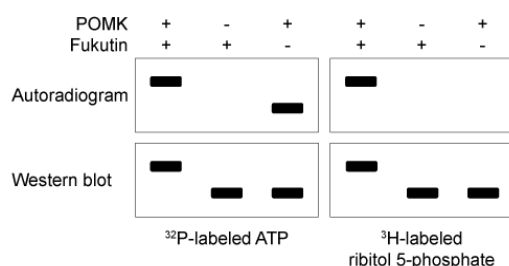


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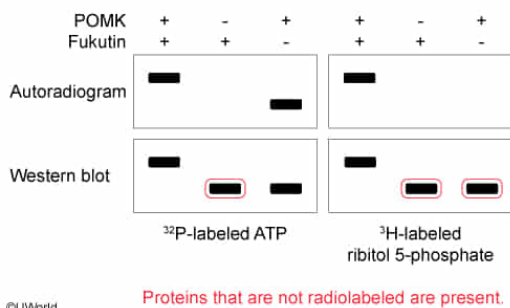
What was the purpose of the western blot in Figure 2?

- ✔ ☒ A. It confirmed the presence of Dg in all lanes (positive control). [64%]
- ☐ B. It showed that the molecular weight of Dg increases when enzymes are absent (dependent variable). [11%]
- ☐ C. It demonstrated experimental conditions for each lane (independent variable). [17%]
- ☐ D. It showed unmodified Dg (negative control). [5%]

Correct

64% answered correctly

Explanation:



Autoradiography only detects molecules that contain radioactive atoms; lanes containing molecules with no radiolabel will appear empty. However, it is also possible that those lanes appear empty because they contain no molecule at all, with or without radiolabels. The **western blot** was performed with an anti-Dg antibody that detects all Dg molecules and confirms that they are present, even if they are not radiolabeled. The researchers can then be sure that the absence of bands in the autoradiogram is due to failure of the relevant enzyme to modify Dg and not due to a problem with isolating Dg itself. Note that unmodified Dg runs further on the gel because the absence of carbohydrates yields a smaller molecule.

Tests of this nature are called **positive controls**, and they help ensure that each step of the experimental procedure performs as expected. The first lane in each autoradiogram is also a positive control, showing that radiolabels are added when all enzymes are present.

(Choice B) The dependent variable in an experiment is the parameter that may change in response to changes in experimental conditions. In this case, whether or not Dg gains a radiolabel (either ³²P or ³H) is the dependent variable. The molecular weight of Dg *decreases* when enzymes are absent.

(Choice C) The independent variable is the experimental condition that is changed by the researchers for each part of the experiment. In this case, the presence or absence of POMK and fukutin is the independent variable.

(Choice D) A negative control shows what the outcome would look like if the results were negative and helps confirm that positive results come from the expected source. In this case, the lane with no POMK in the ³²P experiment and the lane with no fukutin in the ³H experiment are negative controls because it is expected that the radiolabels will not be added when the relevant enzymes are absent.

Educational objective:

A proper experimental approach requires controls to confirm that the results are related to the conditions being tested rather than experimental error or confounding factors. Controls can be

positive or negative, showing what a positive or a negative result should look like, respectively.

Time spent:0 sec

Subject:
Biochemistry

QID:401009

System:
Carbohydrates, Nucleotides, and Lipids